

Georgios Palkanoglou

Current position

Postdoctoral Researcher

University of Jyväskylä, Jyväskylä, Finland

Contact

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Employment

2026–	Postdoctoral Researcher, University of Jyväskylä, Jyväskylä, Finland
2023–2026	Postdoctoral Researcher, TRIUMF, Vancouver, Canada

Education

2019–2023	Ph.D. in Physics, University of Guelph	Dissertation: <i>Pairing in Nuclear and Cold Atomic Systems</i>
2018–2019	M.Sc. in Physics, University of Guelph	Thesis: <i>Approaching the Thermodynamic Limit in BCS Theory</i>
2013–2017	B.Sc. (HONOURS) in Physics, National and Kapodistrian University of Athens	

Awards & Honours

2025–2026	Catalonia Quantum Academy Visiting Scholar Award
2025	Canadian Institute for Nuclear Physics support for workshop organization
2020	Constantine and Patricia Mavroyannis Scholarship
2019–2020	Dean's Graduate scholarship (University of Guelph)
2019	International Doctoral Training Scholarship - Ph.D.'s (University of Guelph)

Conference organizing

2027	(<i>Upcoming</i>) Lead organizer of the ECT* workshop " <i>Bridging Effective Field Theory Across Subatomic Scales</i> "
2025	Lead organizer of the ESNT workshop (with CINP support) " <i>Ab initio many-body calculations: where has the pairing gone?</i> "
2025	Lead organizer of the TRIUMF workshop " <i>SMEFT meets ChEFT</i> "
2025	Co-organizer of the TRIUMF workshop " <i>PAINT26: Progress in Ab Initio Nuclear Theory</i> "

Spokesperson of Collaboration

2025	S1936 at TRIUMF: <i>Beam Development of Light Lanthanides for Nuclear Structure Investigations Approaching $N = Z$</i>
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Teaching experience

2026	Instructor for " <i>Ab initio nuclear theory: a crash course</i> ", University of Barcelona, Barcelona, Spain
2023	Graduate Teaching Assistant in Physics of Music, Department of Physics, University of Guelph
2021	Graduate Teaching Assistant in Quantum Mechanics, Department of Physics, University of Guelph
2020	Graduate Teaching Assistant in (graduate) Quantum Field Theory, Department of Physics, University of Guelph
2019–2023	Private Tutor in Undergraduate Physics
2018–2023	Graduate Teaching Assistant in Applied Physics, Department of Physics, University of Guelph
2013–2017	Private Tutor in Physics, Maths, and Geometry

Student co-Supervision

2026–2024–2025	Isaiah Bennet, M.Sc. student, University of Guelph Project: <i>Emulators for Hartree-Fock-Bogoliubov methods</i>
	Christos Tsekouras, Undergraduate thesis, National and Kapodistrian University of Athens Project: <i>The quantum information in the scattering of nucleons</i>
2021–2022	Michael Stuck, Undergraduate Co-op student, University of Guelph Project: <i>Spin triplet pairing in deformed heavy nuclei</i>
2022	Kareena Bhalla, Undergraduate Co-op student, University of Guelph Project: <i>Non-linear regression for dilute gases of cold atoms</i>

Publications

JOURNAL ARTICLES

- 2026 | *Exploring the shell structure of trapped superfluids*
G. Palkanoglou, R. Curry, and S. Gandolfi
[arXiv:2607.14214](#) [nucl-th]
- 2026 | *Interplay between Nuclear Shell Structure and Pairing around Doubly Magic ^{132}Sn*
R. Simpson, G. Palkanoglou, E.C. Brisley, J.D. Cardona, A. Czihaly, S. Kakkar, M. Simonov, E. Taylor, C. Walls, P. Weligampola, C. Chambers, F. Maldonado Millan, A. Mollaebrahimi, D. Ray, A. Weaver, J. Yu, I. Dillmann, A. Gezerlis, G. Gwinner, A.O. Macchiavelli, S. Malbrunot-Ettenauer, M.P. Reiter, and A.A. Kwiatkowski
[arXiv:2607.05647](#) [nucl-ex]
- 2026 | *Evidence for Multimodal Superfluidity of Neutrons*
Y. Ma, G. Palkanoglou, J. Carlson, S. Gandolfi, A. Gezerlis, G. Given, A. Hicks, D. Lee, K.E. Schmidt, and J. Yu
[arXiv:2602.17611](#) [nucl-th]
- 2025 | *A novel way of recasting the Bardeen-Cooper-Schrieffer gap equations*
G. Palkanoglou and A. Gezerlis
Phys. Rev. C **114**, 024340 (2026) | [arXiv:2510.07384](#) [nucl-th]
- 2025 | *Symmetry properties of pairing correlations in heavy deformed nuclei*
G. Palkanoglou and A. Gezerlis
[arXiv:2505.08879](#) [nucl-th]
- 2024 | *Spin-Triplet Pairing in Heavy Nuclei is Stable Against Deformation*
G. Palkanoglou, M. Stuck, and A. Gezerlis
Phys. Rev. Lett. **134**, 032501 (2025) | [arXiv:2402.13313](#) [nucl-th]
- 2022 | *The 1S_0 pairing gap in neutron matter*
S. Gandolfi, G. Palkanoglou, J. Carlson, A. Gezerlis, and K. E. Schmidt
Condens. Matter **2022**, 7(1), 19 | [arXiv:2201.01308](#) [nucl-th]
*Marked as a Feature Paper
- 2021 | *Superfluid neutron matter with a twist*
G. Palkanoglou and A. Gezerlis
Universe **2021**, 7(2), 24 | [arXiv:2012.04663](#) [nucl-th]
* Marked as a Feature Paper
- 2020 | *From odd-even staggering to the pairing gap in neutron matter*
G. Palkanoglou, F. K. Diakonou, and A. Gezerlis
Phys. Rev. C **102**, 064324 (2020) | [arXiv:2005.05985](#) [nucl-th]

PROCEEDINGS

- 2019 | *Effective mass and pairing gap in neutron matter*
G. Palkanoglou, N. Ismail, M. Buraczynski, and A. Gezerlis
J. Phys.: Conf. Ser. **1643**, 012132 (2020).

Invited Talks & Seminars

- 2026 | *Beyond the conventional nuclear superfluidity: triplet and mixed-spin superfluids across nucleonic systems, pairing at the drip-line, and more*

	University of Barcelona, Barcelona, Spain
2026	<i>Revisiting superfluidity and shell-effects at the crossroads: experiments, mean-field, and ab initio descriptions</i> UPC Universitat Politècnica de Catalunya, Barcelona, Spain
2026	<i>Nuclear superfluidity and clustering, many-nucleon forces, and ML methods for ab initio nuclear theory</i> Job Interview (virtual), ECT*, Trento, Italy
2026	<i>Unconventional nuclear superfluidity: from the proton to the neutron drip-line, and beyond</i> Job Interview (virtual), University of Jyväskylä, Jyväskylä, Finland
2026	(Co-organized) <i>Beyond the leading three-nucleon force in light nuclei</i> PAINT ₂₆ , Vancouver, BC, Canada
2025	(Organized) <i>Pinning the subleading three-nucleon contact interaction to light nuclei</i> SMEFT meets ChEFT workshop, TRIUMF, Vancouver, BC, Canada
2025	(Organized) <i>Ab initio calculations of singlet and triplet pairing gaps in nucleonic matter</i> Introductory lecture at Ab initio many-body calculations: “where has the pairing gone?”, ESNT CEA, Saclay, Paris, France
2025	<i>How to describe isoscalar pairing in heavy deformed nuclei and measure its effects</i> ACOT meeting, TRIUMF, Vancouver, BC, Canada
2025	<i>Putting the subleading three nucleon force to use</i> PAINT ₂₅ , TRIUMF, Vancouver, BC, Canada
2025	<i>S1936 Progress Report: Beam Development of Light Lanthanides for Nuclear Structure Investigations Approaching $N = Z$</i> NP-EEC, TRIUMF, Vancouver, BC, Canada
2024	<i>Exotic pairing phases in nuclei and neutron matter</i> T-2 Seminar, LANL, Los Alamos, NM, USA
2024	<i>Exotic nuclear superfluidity in heavy nuclei</i> 3rd APCTP-TRIUMF joint workshop, PKNU, Busan, South Korea
2024	<i>Nuclear superfluidity in nuclei and neutron matter</i> National and Kapodistrian University of Athens, Athens, Greece
2024	(Pathways to) <i>Exotic pairing in heavy nuclei</i> PAINT ₂₄ - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, BC, Canada
2023	<i>Novel pairing and testing EFT interactions in neutron matter and nuclei</i> Michigan State University
2023	<i>Mixed-spin pairing in nuclear and cold atomic systems</i> (Virtual) Warsaw University of Technology (Job Interview)
2023	<i>Mixed-spin pairing in nuclear and cold atomic systems</i> (Virtual) TRIUMF Theory Group (Job Interview)
2023	<i>Representing sound: Acoustics & Phonetics</i> One lecture in Phonetics School of Languages and Literature, University of Guelph, Guelph, ON, Canada

Contributed talks

2026	<i>Superfluid effects in deformed heavy nuclei close to the drip-lines</i> Nuclear Theory in Spain I, Universidad Complutense de Madrid, Madrid, Spain
2024	<i>Exotic nuclear superfluidity in heavy nuclei</i> 2024 CeNAM Frontiers, University of Notre Dame, South Bend, IN, USA
2024	<i>Exotic nuclear superfluidity in heavy nuclei</i> Theory Canada 16, Western University, London, ON, Canada
2024	<i>Exotic nuclear superfluidity in heavy nuclei</i> CAP Congress 2024, Western University, London, ON, Canada
2023	<i>Singlet and triplet pairing in nuclear and cold atomic systems</i>

	2023 Canadian Association of Physicists Congress University of New Brunswick, Fredericton, NB, Canada
2023	<i>Singlet and triplet pairing in nuclear and cold atomic systems</i> 60 th Winter Nuclear and Particle Physics Conference Banff, AB, Canada
2023	<i>Mixed-spin pairing in nuclear and cold atomic systems</i> Toronto Ultracold Atom Network (TUCAN) meeting University of Toronto, Toronto, ON, Canada
2022	<i>The S-wave pairing gap in neutron matter</i> 2022 Canadian Association of Physicists Congress *Marked as the 3 rd best presentation in the Division of Theoretical Physics (DTP)
2021	<i>Superfluid neutrons: from particles to matter</i> 2021 Canadian Association of Physicists (Virtual) Congress *Tied as the 2 nd best presentation in the Division of Theoretical Physics (DTP)
2021	<i>Superfluid neutron matter with a twist</i> 58 th Winter Nuclear and Particle Physics (Virtual) Conference * Marked as the 2 nd best presentation in Theoretical Physics
2021	<i>Superfluid neutron matter with a twist</i> (Virtual) Graduate Seminar Series University of Guelph, Guelph, ON, Canada
2020	<i>From odd-even staggering to the pairing gap in neutron matter</i> 2020 Canadian Association of Physicists (Virtual) Congress
2020	<i>From odd-even staggering to the pairing gap in neutron matter</i> 57 th Winter Nuclear and Particle Physics Conference Banff, AB, Canada
2018	<i>A Superconducting Star</i> Graduate Seminar Series University of Guelph, Guelph, ON, Canada

Conferences & Summer schools attended

2026	ML2026 - II Machine Learning Methods for Complex and Quantum Systems, University of Barcelona, Barcelona, Spain
2026	NT.ES I - Nuclear Theory in Spain, Universidad Complutense de Madrid, Madrid, Spain
2026	CSQCD - Compact Stars in the QCD phase diagram, University of Barcelona, Barcelona, Spain
2026	PAINT26 - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada
2025	SMEFT meets ChEFT, TRIUMF, Vancouver, British Columbia, Canada
2025	Ab initio many-body calculations: where has the pairing gone, ESNT, Saclay, France
2025	PAINT25 - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada
2024	3rd APCTP-TRIUMF joint workshop: From Nuclei to Neutron Stars, PKNU, Busan, South Korea
2024	2024 CeNAM Frontiers, University of Notre Dame, South Bend, Indiana, USA
2024	2024 C.A.P Congress, Western University, London, Ontario, Canada
2024	Theory Canada 16, Western University, London, Ontario, Canada
2024	PAINT24 - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada
2023	2023 C.A.P Congress, University of New Brunswick, Fredericton, New Brunswick, Canada
2023	60 th Winter Nuclear and Particle Physics (Virtual) Conference, Banff, AB, Canada
2022	2022 C.A.P Congress, McMaster University, Hamilton, Ontario, Canada
2021	2021 C.A.P (Virtual) Congress
2021	58 th Winter Nuclear and Particle Physics (Virtual) Conference
2020	2020 C.A.P (Virtual) Congress
2020	57 th Winter Nuclear and Particle Physics Conference, Banff, AB, Canada
2017	The 2017 Onassis Foundation Lectures in Physics: "Quantum physics frontiers explored with cold atoms, molecules and photons", Heraklion, Greece
2017	The 2017 Petnica Summer Institute summer school on Cosmology, Petnica, Serbia
2016	The 3 rd Xmas Theoretical Physics Workshop, Athens, Greece

Computing skills

LANGUAGES

Fortran, Python, Bash, MATLAB, Mathematica, C++, HTML

SOFTWARE

L^AT_EX, PLUTO, Logger Pro

Outreach & Service

2025	Volunteer mentor in the Early Career Researcher mentorship program, TRIUMF
2022	Research Information Session, University of Guelph
2021	Volunteer judge for the 57th Annual Canadian Undergraduate Physics Conference (CUPC)
2019	Volunteer for STEM Week 2019 organized by the University of Guelph, Guelph Civic Museum
2018	Volunteer for STEM Week 2018 organized by the University of Guelph, Guelph Civic Museum

Languages

Greek (native), English (fluent), Farsi (beginner), Spanish (beginner), German (beginner)